A Benchmarked Pipeline for Ferroptosis-Associated Prognosis, Classification and Drug-Response Analysis in Colon Cancer: Evidence Against Gene-Set Specificity

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Abstract

Ferroptosis, an iron-dependent form of regulated cell death, is widely proposed as a source of prognostic and predictive biomarkers in colorectal cancer. Reported ferroptosis signatures are rarely compared with equally sized random gene sets, so apparent performance may reflect generic transcriptomic signal rather than ferroptosis biology. This study presents a fully scripted pipeline that applies one design rule to every analysis: each result obtained with a 36-gene ferroptosis panel is compared with 200 random gene sets of the same size that pass through the identical procedure. The pipeline covers (i) a ridge-penalised Cox prognosis model trained on TCGA-COAD (n = 434 with valid overall survival) and frozen before evaluation in three independent GEO cohorts; (ii) classification of mismatch-repair deficiency (dMMR); (iii) unsupervised ferroptosis subtypes; and (iv) prediction of cell-line GPX4 dependence and chemotherapy IC_{50} from expression. In external validation the frozen model gave concordance indices of 0.539 (GSE39582), 0.486 (GSE17536) and 0.581 (GSE14333), none distinguishable from random gene sets (empirical p = 0.338, 0.831, 0.343), and below a clinical model of stage and age (0.675 and 0.747 in the two OS cohorts). The frozen score was not significantly associated with outcome in any single cohort (pooled OS hazard ratio per SD 1.09, 95% CI 0.97 to 1.22), and its increment in C-index over age and stage was between −0.022 and +0.018. dMMR was classified with AUC 0.958 but random gene sets reached a median near 0.94 (empirical p = 0.234). The panel predicted GPX4 dependence in 59 colorectal cell lines with AUC 0.760, again at the random-set median (0.776). Only one of five response models, oxaliplatin IC_{50}, outperformed random sets nominally (p = 0.045), and not after multiplicity correction (q = 0.22). Compared with ferroptosis signatures published since 2023, whose reported external 5-year AUCs fell to 0.63–0.65 from 0.74–0.97 in discovery cohorts, the frozen model reached 0.553 (GSE39582), consistent with a pervasive optimism gap. These findings show that, for this panel, apparent predictive performance does not demonstrate ferroptosis-specific information, and they argue for random-gene-set benchmarking as a default control in signature studies.

Keywords: Ferroptosis; colorectal cancer; prognostic signature; random gene sets; external validation; drug response; reproducibility.

1 Introduction

Colorectal cancer remains a leading cause of cancer-related mortality worldwide, and molecular markers that improve prognosis or treatment selection beyond stage are of continuing interest. Ferroptosis is a form of regulated cell death driven by iron-dependent lipid peroxidation and restrained by the glutathione–GPX4 axis and parallel suppressor pathways [1]. Genes in iron handling, polyunsaturated lipid metabolism and antioxidant defence have therefore been proposed as features for prognostic models and as determinants of drug sensitivity. Cell state, notably a mesenchymal phenotype, has been linked to GPX4 dependence [2, 3].

A recurring weakness in gene-signature studies is the lack of an appropriate null. A model built on any gene set of reasonable size may show non-trivial survival association, because proliferation, stromal content and tumour stage affect the expression of thousands of genes. Without comparison with random gene sets processed identically, a modest C-index or AUC cannot be attributed to the biology that motivated the gene list.

This work develops an end-to-end computational pipeline for ferroptosis-related analysis in colon cancer in which that comparison is built in. The objectives were:

1. to train a gene-expression survival model in TCGA-COAD and freeze it before testing in independent cohorts;
2. to assess whether the panel carries information about mismatch-repair status and defines survival-relevant subtypes;
3. to assess whether the panel predicts GPX4 dependence

and drug response in colorectal cell lines; and
4. to judge every result against equally sized random gene sets, and against positive controls that show whether the machinery can detect a real signal.

2 Related Work

Numerous studies have reported ferroptosis-related signatures with prognostic value in colorectal and other cancers, typically derived by univariate screening followed by penalised Cox regression. Performance is often reported in the discovery cohort or in a single validation set, and rarely against random-gene-set baselines. Methodological work has shown that many random gene sets are associated with outcome [16], and that feature selection outside the cross-validation loop inflates performance. Large pharmacogenomic resources [5, 6] and the DepMap project [7] enable cell-line analyses, but the correspondence between cell-line response and patient outcome is limited.

Since 2023, compact colorectal models have continued to appear. Examples include a four-gene LASSO Cox score combined with clinical factors that reached AUCs of 0.897 in TCGA and 0.858 in GSE39582 [19], a multi-model study of random survival forests and other learners on six intersection genes [12], and a mitochondrial ferroptosis score validated in GSE39582 [13]. A 2026 pan-cancer benchmark is a notable exception, because it showed that an integrated cell-death signature exceeded size-matched random gene sets but did not outperform a ferroptosis-only model [23]. This study therefore emphasises out-of-fold estimation, frozen models, empirical comparison with random sets and explicit reporting of negative results. A structured, quantitative comparison with the studies published from 2023 onwards is given in Section 4.10.

3 Materials and Methods

3.1 Overview

Figure 2 summarises the workflow. The code is a single Python script (version 1.5.0, fixed seed 42) that downloads data, preprocesses them, runs all analyses and writes tables, figures and a provenance manifest. Nothing is simulated: an analysis whose data cannot be retrieved is skipped and reported as such. Figure 1 shows how each score and classifier is computed from the standardised panel expression.

3.2 Datasets

Table 1 lists the data used. TCGA-COAD expression (STAR gene-level counts, GDC Data Release 46.0) was converted to $\log_2(\mathrm{TPM}+1)$ for protein-coding genes. One primary-tumour sample per patient was kept (524 files reduced to 458 patients). Overall survival (OS) was defined from vital status, days to death or last follow-up, with 434 patients valid (time > 0, event known; 95 deaths). The GEO cohorts (GSE39582 [4], GSE17536, GSE14333) are Affymetrix GPL570 microarrays; probes were collapsed to genes by retaining the probe with the highest mean expression and discarding probes mapping to several genes. GSE14333 provides no OS, so disease-free survival (DFS) was used (exploratory, endpoint mismatch with the OS-trained model). Pairwise correlation of samples between GEO cohorts identified 95 GSE14333 samples that duplicate GSE17536 ($r > 0.9$); they were removed, leaving 153 patients with DFS and 29 events.

Table 1: Cohorts after quality control. Events are deaths (OS) or recurrences (DFS).

Cohort	Type	Endpoint	n	Events
TCGA-COAD	RNA-seq	OS	434	95
GSE39582	GPL570	OS	573	190
GSE17536	GPL570	OS	177	73
GSE14333	GPL570	DFS	153	29
DepMap/GDSC2	cell lines	see text	39–59	–

For cell-line analyses, expression was taken from DepMap (protein-coding TPM, $\log_2(\mathrm{TPM}+1)$). The download of the GDSC RMA expression file and of CTRPv2 failed in the final run, so CTRP was not analysed; GDSC2 fitted dose–response values (colorectal lineage) were matched to DepMap lines by COSMIC identifier or normalised name. Genetic GPX4 dependence was taken from the DepMap CRISPR gene-effect score for 59 colorectal (bowel-lineage) lines with expression and score. No GPX4-inhibitor (ML162, ML210, RSL3) or erastin data were present in the retrieved GDSC2 file.

3.3 Ferroptosis Gene Panel

The panel comprises 36 genes (16 drivers, 20 suppressors; Appendix Table 11) compiled from the general ferroptosis literature. It is a candidate list that has not been independently curated against a ferroptosis database. All findings are therefore conditional on this panel (hash 03d5812616df).

3.4 Preprocessing and Harmonisation

RNA-seq and microarray values are never pooled. In each external cohort the selected genes are z-scored within the cohort using no outcome information and multiplied by coefficients frozen in TCGA. No validation outcome is used for selection, tuning or thresholds.

3.5 Survival Risk Model

The Cox model [10] is

$$h(t;\mathbf{x}) = h_0(t)\exp\left(\boldsymbol{\beta}^T\mathbf{x}\right), \tag{1}$$

and the risk score for patient i is

$$R_i = \sum_{j=1}^{p}\beta_j z_{ij}, \tag{2}$$

where z_{ij} is the standardised expression of selected gene j. In TCGA, genes are standardised, screened by univariate Cox ($p < 0.10$), and a ridge-penalised Cox model is fitted with the penalty chosen from $\{0.05, 0.5, 5, 50, 500\}$ by 3-fold inner cross-validation of the C-index. Three genes were selected (ALOX12, AKR1C1, HSPB1) with penalty 5. The model and the median risk cut-off were frozen before external evaluation. TCGA cross-validation (5-fold, 2 repeats) repeats screening, tuning and fitting inside each training fold. A clinical model

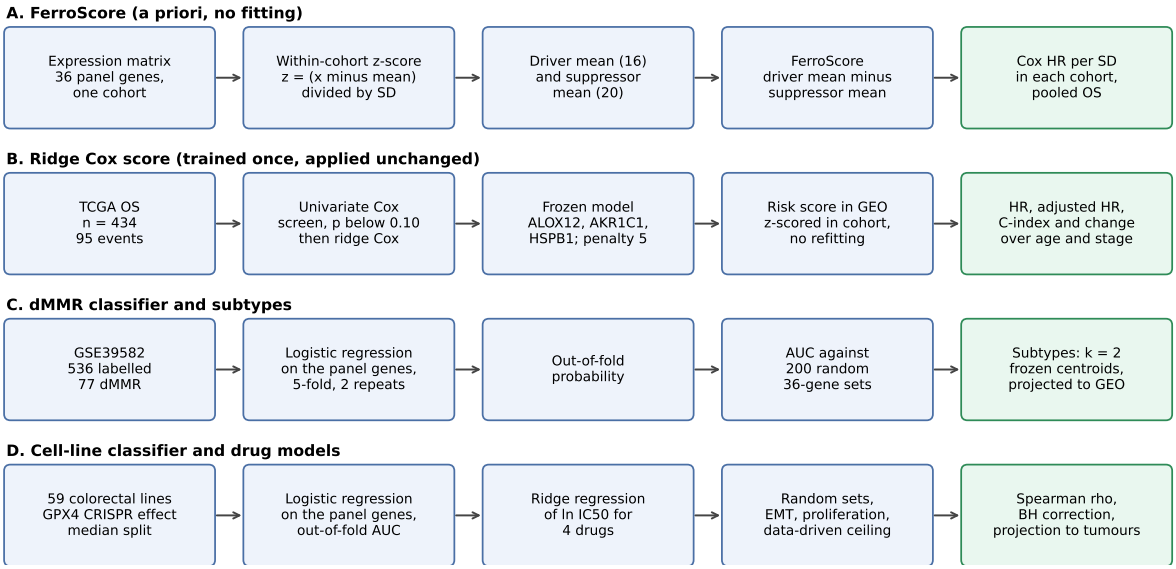


Figure 1: Method-level architecture. After within-cohort standardisation the panel feeds (A) the a priori FerroScore, (B) the ridge Cox score trained once on TCGA overall survival and applied unchanged to the GEO cohorts, (C) the dMMR classifier and the ferroptosis subtypes, and (D) the cell-line classifier and drug models. Green boxes are the outputs that are judged against baselines.

(age, stage) and a combined model (age, stage, risk) were fitted in TCGA and frozen likewise. Discrimination is Harrell's C-index [8] with bootstrap 95% confidence intervals (500 resamples).

3.6 Random-Gene-Set Benchmark

For every task, 200 random gene sets of the same size as the panel (36 genes) were drawn without replacement from expressed genes shared by all cohorts, excluding panel genes. Each set passes through the *identical* procedure (screening, tuning, fitting, frozen application). The empirical one-sided p-value is

$$p_{\text{emp}} = \frac{1 + \#\{c_k \geq c_{\text{obs}}\}}{1 + N}, \tag{3}$$

with $N = 200$ and c the relevant metric. For subtype survival tests, the absolute log hazard ratio is compared. A p_{emp} near or above 0.05 means the panel cannot be distinguished from arbitrary genes for that task. Because p_{emp} cannot fall below one in 201, it is a resolution-limited statistic.

3.7 Molecular Classification

dMMR (1) versus pMMR (0) labels were parsed from GSE39582 (536 patients with labels, 77 dMMR, 14.4%). An L2 logistic regression,

$$P(Y = 1; \mathbf{x}) = \frac{1}{1 + \exp[-(\beta_0 + \boldsymbol{\beta}^T \mathbf{x})]}, \tag{4}$$

was evaluated by patient-level out-of-fold (OOF) prediction (5-fold, 2 repeats) with in-fold standardisation. A 12-gene proliferation marker set served as a comparator. TCGA MSI labels were not retrieved.

3.8 Ferroptosis Subtypes

k-means with $k = 2$ (prespecified) was fitted on TCGA z-scores of the panel; the cluster with the higher mean driver-minus-suppressor score was labelled "high". Centroids were frozen, and each GEO cohort was assigned by nearest centroid after cohort-wise z-scoring. Subtypes were tested against survival (log-rank, Cox with and without age and stage), stage, age and MMR, and compared with random sets by the same clustering-and-projection procedure.

3.9 Cell-Line Sensitivity and Drug-Response Prediction

Colorectal cell lines were labelled sensitive or resistant to GPX4 loss by the median split of the CRISPR gene-effect score (29 sensitive, 30 resistant). The panel-based logistic classifier was evaluated by OOF AUC (5-fold, 3 repeats), a label-permutation test (200 permutations) and the random-set benchmark. The frozen classifier was then projected onto the tumour cohorts. For dose-response, ridge regression

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left[\sum_{i=1}^{n} (y_i - \mathbf{x}_i^T \boldsymbol{\beta})^2 + \lambda \sum_j \beta_j^2 \right] \tag{5}$$

predicted ln IC_{50} (μM) of 5-fluorouracil, irinotecan, oxaliplatin and SN-38, and the CRISPR gene effect, from panel expression (y_i the response of line i). Performance is the Spearman correlation ρ between OOF predictions and measurements, judged against random sets, with Benjamini–Hochberg (BH) [9] correction across the five models. Out-of-fold ρ is biased low for pure noise, so only the comparison with random sets is interpreted.

3.10 Positive Controls

To test whether the machinery can detect real signal, three references were run identically: a mesenchymal and epithelial (EMT) signature (24 genes) and a proliferation signature (12 genes) for GPX4 dependence and survival; clinical stage for survival; and a data-driven ceiling in which the 36 most

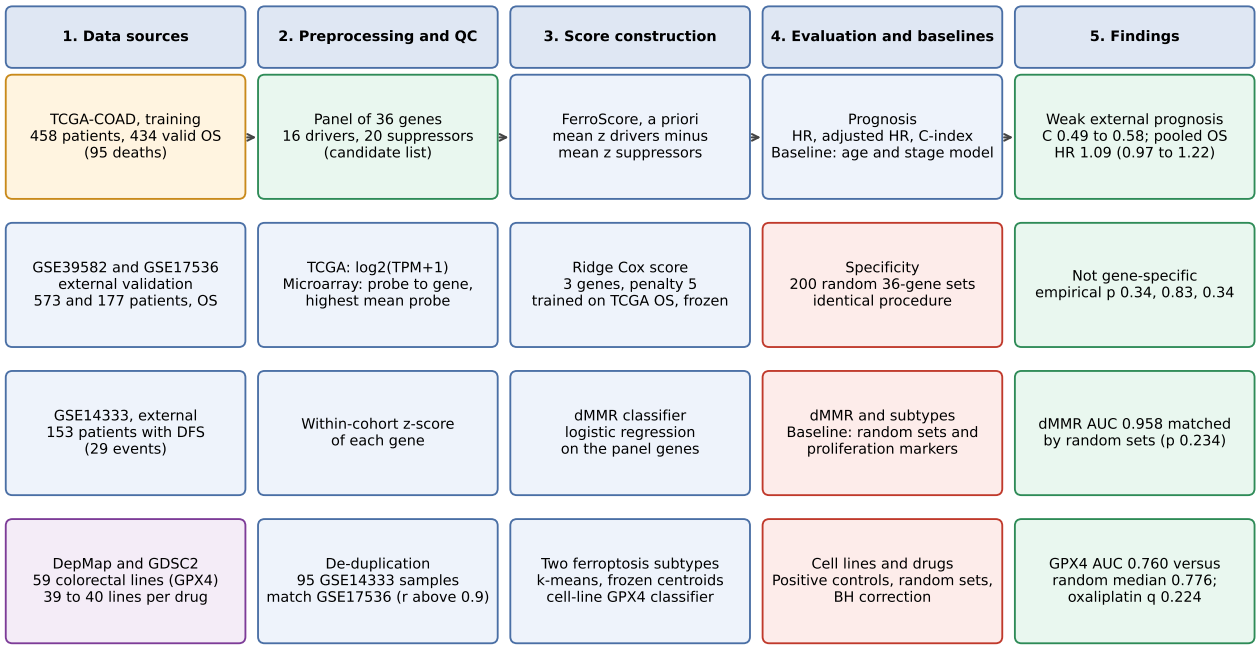


Figure 2: Architecture of the study. Public colorectal cancer and cell-line data are harmonised (stages 1 and 2), the scores and classifiers are constructed (stage 3), and each evaluation module is paired with a baseline or null model (stage 4, shaded red) before the findings are drawn (stage 5). Counts are those retained after quality control.

discriminative genes are chosen inside each training fold. Cell-line results describe in vitro response and are not clinical doses.

3.11 Evaluation Metrics, Adjustment and Pooling

Within each external cohort the frozen risk score was standardised, and the hazard ratio per standard deviation (SD) was estimated by Cox regression. Adjusted models included ordinal stage (TNM or AJCC stage in GSE39582 and GSE17536, Dukes A to D coded 1 to 4 in GSE14333) and age per 10 years, using complete cases. The proportional-hazards assumption was tested with scaled Schoenfeld residuals. Incremental discrimination was defined as

$$\Delta C = C_{\text{age+stage+risk}} - C_{\text{age+stage}}, \tag{6}$$

where both models carry coefficients frozen in TCGA, and the 95% interval was obtained from 1,000 bootstrap resamples of the same patients. Log-rank tests used the frozen TCGA median of the risk score, and risk tertiles used the frozen TCGA tertile cut-offs. Cohort-level log hazard ratios were combined by inverse-variance fixed-effect and DerSimonian–Laird random-effects models, and heterogeneity was summarised by I^2. Pooling was done within the overall-survival endpoint, because mixing endpoints is not interpretable.

A priori FerroScore. To test the ferroptosis genes without any fitting, a score was defined from the roles assigned to the panel genes before any survival analysis:

$$\text{FerroScore}_i = \frac{1}{n_D}\sum_{j\in D} z_{ij} - \frac{1}{n_S}\sum_{j\in S} z_{ij}, \qquad z_{ij} = \frac{x_{ij}-\mu_j}{\sigma_j}, \tag{7}$$

where D and S are the 16 driver and 20 suppressor genes ($n_D = 16$, $n_S = 20$), and μ_j and σ_j are the mean and standard deviation of gene j within the cohort. The driver mean and the suppressor mean were also analysed separately. This component analysis was added after the main results were seen and is exploratory.

3.12 Baseline Comparison Protocol

Published ferroptosis signatures were used as external baselines through a fixed, four-step protocol, so that the comparison does not depend on selective reading.

Step 1, identification. Studies were included if they were published from 2023 onwards, built a ferroptosis-related gene model for colon or colorectal cancer from TCGA and GEO data, and reported at least one discrimination or hazard metric. One pan-cancer benchmark that applied a random-gene null was added because it tests the same question as this study. The search was targeted and not systematic. Studies published before 2023 were used only for context.

Step 2, extraction. For every study the following fields were recorded: year, signature size and model type, discovery and validation cohorts, reported performance, and whether a baseline or a null model was described. Entries were read

from abstracts, methods text and result summaries, and are marked in the table notes when only the abstract was available.

Step 3, comparison rules. Published models were not refitted, because gene lists, coefficients and preprocessing differ. The comparison therefore concerns design and reported metrics. Metrics of the same type (AUC with AUC, *C*-index with *C*-index, hazard ratio with hazard ratio) are placed side by side, and metrics of different types are listed but not ranked. Each pair receives a comparability grade: *high* when the metric type and the external cohort are the same, *moderate* when the metric type is the same but the cohort or cancer scope differs, or when both values are in-sample, and *low* when the metric type differs or the published value belongs to a model combined with clinical variables.

Step 4, a comparable metric for the frozen model. Published models mostly report time-dependent ROC AUC, whereas the pipeline reports Harrell's *C*. To allow a same-type comparison, the inverse-probability-of-censoring-weighted cumulative and dynamic AUC (Uno-type estimator) was computed for the frozen risk score at 36 and 60 months, with 300 bootstrap resamples for intervals. Estimator details differ between studies, so differences of a few hundredths are not interpreted.

Internal baselines. Independently of the literature, every analysis in this study is paired with a baseline or null model that answers a specific question (Table 10): a clinical stage and age model for prognosis, 200 random gene sets for specificity, and positive controls for the ability to detect a real signal.

4 Results

4.1 Cohort Characteristics

After quality control the four tumour cohorts contained 434, 573, 177 and 153 evaluable patients (Table 1). In TCGA, median follow-up was 22.1 months, median age 68.5 years, and 19.4% of stage values were missing. The panel was fully present in all cohorts (36 of 36 genes).

4.2 Survival Model Performance

Univariate screening in TCGA retained ALOX12 (HR 1.40 per SD, 95% CI 1.18–1.65, BH $q = 0.005$), AKR1C1 (HR 1.23) and HSPB1 (HR 1.23); only ALOX12 survived multiplicity correction. TCGA cross-validation with selection inside folds gave $C = 0.548$ for the molecular model, against 0.770 for the clinical model and 0.774 for the combined model (in-sample molecular $C = 0.617$, optimistic). Table 2 gives the frozen external results. The molecular model was close to chance in all cohorts, and the clinical model was clearly superior in the two OS cohorts (Fig. 3; risk-group survival curves are shown in Fig. 4). Adding the molecular risk score to the clinical model changed *C* from 0.675 to 0.683 in GSE39582 and from 0.747 to 0.725 in GSE17536 (Fig. 11).

Table 2: Frozen external validation (Harrell *C*, 95% bootstrap CI) and comparison with 200 random gene sets.

Cohort	Molecular *C*	Random median	p_{emp}	Clinical *C*
GSE39582	0.539 [0.499, 0.580]	0.528	0.338	0.675
GSE17536	0.486 [0.415, 0.562]	0.532	0.831	0.747
GSE14333†	0.581 [0.471, 0.701]	0.555	0.343	0.624

†DFS endpoint, 29 events; exploratory.

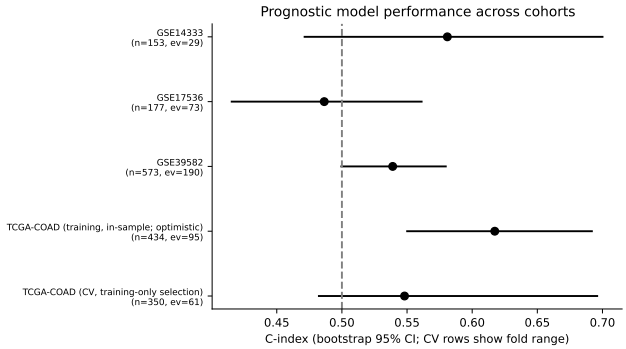


Figure 3: Concordance index of the molecular model: TCGA cross-validation (training-only selection, interval shows fold range), TCGA in-sample, and frozen external cohorts.

4.3 Adjusted Association and Incremental Value

Unadjusted, the frozen score was not significantly associated with outcome in any external cohort (Table 3, Fig. 5). The hazard ratio per SD was 1.13 (95% CI 0.99 to 1.29, $p = 0.080$) for OS in GSE39582, 0.97 (0.76 to 1.23, $p = 0.783$) for OS in GSE17536 and 1.34 (0.93 to 1.93, $p = 0.120$) for DFS in GSE14333. The proportional-hazards assumption was not rejected in any cohort (Schoenfeld p between 0.29 and 0.74). The fixed-effect pooled OS estimate was 1.09 (0.97 to 1.22, $p = 0.164$), with low heterogeneity ($I^2 = 0.17$).

After adjustment for age and stage, the estimate became significant in GSE39582 (HR 1.20, 1.04 to 1.37, $p = 0.011$) but reversed direction in GSE17536 (HR 0.75, 0.58 to 0.98, $p = 0.035$), and remained non-significant for DFS in GSE14333 (HR 1.24, 0.87 to 1.76, $p = 0.230$). The two OS cohorts therefore disagree after adjustment ($I^2 = 0.89$), the random-effects pooled estimate was 0.96 (0.61 to 1.51), and the adjusted fixed-effect pooled value (1.08, 0.96 to 1.22) should not be interpreted. Because the unadjusted estimates in GSE17536 are close to 1 while stage is strongly prognostic there, the reversal is most likely a statistical consequence of conditioning on stage in a cohort of 177 patients, and not evidence of a protective effect.

Incremental discrimination over age and stage was small or negative (Table 4). The change in *C*-index was +0.008 (95% bootstrap interval −0.004 to +0.022) in GSE39582, −0.022 (−0.039 to −0.003) in GSE17536, where adding the score lowered discrimination, and +0.018 (−0.018 to +0.057) in GSE14333. In GSE39582, the frozen TCGA tertiles gave a monotonic but marginal pattern: relative to the lowest tertile, the middle tertile had HR 1.41 (0.98 to 2.02, $p = 0.061$) and the highest tertile 1.47 (1.00 to 2.15, $p = 0.047$). The continuous score remains the primary analysis, and the tertile result is exploratory.

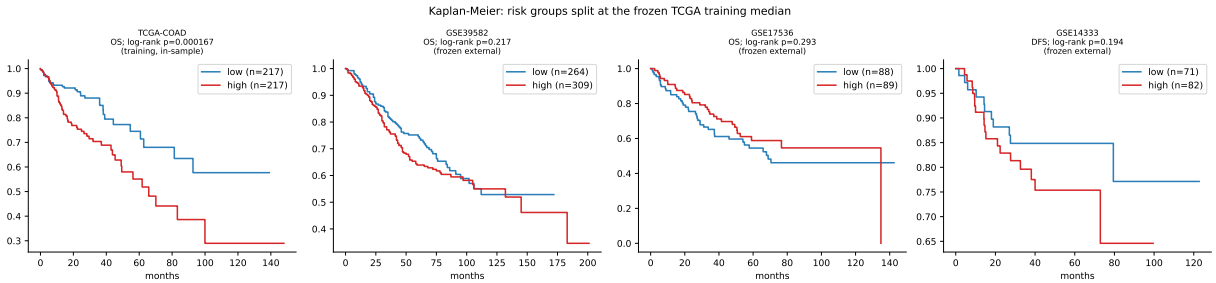


Figure 4: Kaplan–Meier curves for risk groups split at the frozen TCGA median risk, in the training cohort (in-sample) and the external cohorts.

Table 3: External association of the frozen three-gene score with survival, unadjusted and adjusted for age (per 10 years) and stage. Hazard ratios (HR) are per SD of the score within each cohort. PH p is the Schoenfeld-residual test of the proportional-hazards assumption. Adjusted models use complete cases (n = 568, 177 and 153).

Cohort	Endpoint	n	Events	Unadjusted HR (95% CI)	p	PH p	Adjusted HR (95% CI)	p
GSE39582	OS	573	190	1.13 (0.99 to 1.29)	0.080	0.29	1.20 (1.04 to 1.37)	0.011
GSE17536	OS	177	73	0.97 (0.76 to 1.23)	0.783	0.74	0.75 (0.58 to 0.98)	0.035
GSE14333	DFS	153	29	1.34 (0.93 to 1.93)	0.120	0.55	1.24 (0.87 to 1.76)	0.230

Table 4: Discrimination of the frozen score alone, of the frozen age and stage model, and of the combined model (Harrell C, complete cases), with the increment ΔC (bootstrap 95% interval) and the log-rank p for the median split.

Cohort	End.	Score	Age+stage	Combined	ΔC (95% CI)	Log-rank p
GSE39582	OS	0.537	0.675	0.683	+0.008 (−0.004 to +0.022)	0.217
GSE17536	OS	0.486	0.747	0.725	−0.022 (−0.039 to −0.003)	0.293
GSE14333	DFS	0.581	0.624	0.642	+0.018 (−0.018 to +0.057)	0.194

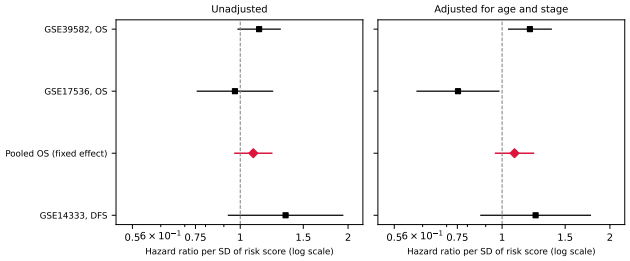


Figure 5: Hazard ratio per SD of the frozen score in the external cohorts. Squares are single cohorts and the red diamond is the fixed-effect pooled estimate for OS. Left, unadjusted; right, adjusted for age and stage.

4.4 A Priori FerroScore and Its Components

The a priori FerroScore, which involves no fitting, was not associated with survival in any cohort (Table 5). Hazard ratios per SD were 1.07 in TCGA-COAD, 0.93 in GSE39582, 0.88 in GSE17536 and 1.05 for DFS in GSE14333, with a fixed-effect pooled OS estimate of 0.95 (0.86 to 1.05, $p = 0.331$, $I^2 = 0$). In the exploratory component analysis, the only nominally significant result was a protective association of the driver mean in GSE39582 (HR 0.80, 0.70 to 0.93, $p = 0.003$). It was not replicated in TCGA-COAD (1.13) or GSE17536 (1.03), the pooled OS estimate was 0.92 (0.83 to 1.03) with $I^2 = 0.76$, and it is one of twelve unadjusted tests. The suppressor mean had a pooled OS hazard ratio of 0.99 (0.90 to 1.10). The ferroptosis role assignment therefore carries no demonstrable prognostic information in these cohorts.

4.5 Specificity Against Random Gene Sets

In none of the three external cohorts did the panel outperform random sets (Fig. 6, Table 2); the 95th percentile of random-set C was 0.575, 0.594 and 0.646, above the observed value in each. The proliferation marker set reached $C = 0.570$ in GSE39582, but 0.498 and 0.509 in the other cohorts, so no gene set was consistently prognostic.

4.6 Classification of dMMR

The panel classified dMMR with OOF AUC 0.958 (95% CI 0.932–0.976; balanced accuracy 0.884; recall 0.857; specificity 0.911; precision 0.617 at prevalence 14.4%) (Fig. 7). The proliferation set reached 0.934 (0.899–0.962). Random gene sets, however, achieved similarly high AUCs, and the panel was not distinguishable from them ($p_{\text{emp}} = 0.234$). The unsupervised driver-minus-suppressor score had AUC 0.453, i.e. no informative orientation. The high AUC therefore reflects the strong, genome-wide expression signature of dMMR rather than ferroptosis biology.

4.7 Ferroptosis Subtypes

The two-cluster solution was only moderately stable (bootstrap adjusted Rand index 0.55; silhouette 0.19 for $k = 2$, declining for larger k). High-ferroptosis tumours showed no significant survival difference in TCGA (HR 0.87, log-rank $p = 0.643$), GSE39582 (HR 1.11, $p = 0.637$), GSE17536 (HR 0.52, $p = 0.099$) or GSE14333 (HR 0.45, $p = 0.424$) (Fig. 8); adjustment for age and stage did not change this. Compared with random sets, the empirical p-values for the absolute

Table 5: A priori FerroScore and its components (HR per SD, 95% CI, p) in the tumour cohorts. TCGA-COAD values are not external. The component analysis is exploratory.

Cohort	Endpoint	n	FerroScore	Driver mean	Suppressor mean
TCGA-COAD	OS	434	1.07 (0.88 to 1.30), 0.516	1.13 (0.92 to 1.38), 0.243	1.05 (0.86 to 1.29), 0.632
GSE39582	OS	573	0.93 (0.81 to 1.07), 0.317	0.80 (0.70 to 0.93), 0.003	0.92 (0.80 to 1.06), 0.229
GSE17536	OS	177	0.88 (0.71 to 1.08), 0.220	1.03 (0.81 to 1.29), 0.825	1.14 (0.91 to 1.43), 0.239
GSE14333	DFS	153	1.05 (0.73 to 1.52), 0.785	1.06 (0.74 to 1.53), 0.751	0.99 (0.69 to 1.44), 0.977
Pooled OS, fixed effect	OS	3 cohorts	0.95 (0.86 to 1.05)	0.92 (0.83 to 1.02)	0.99 (0.90 to 1.10)

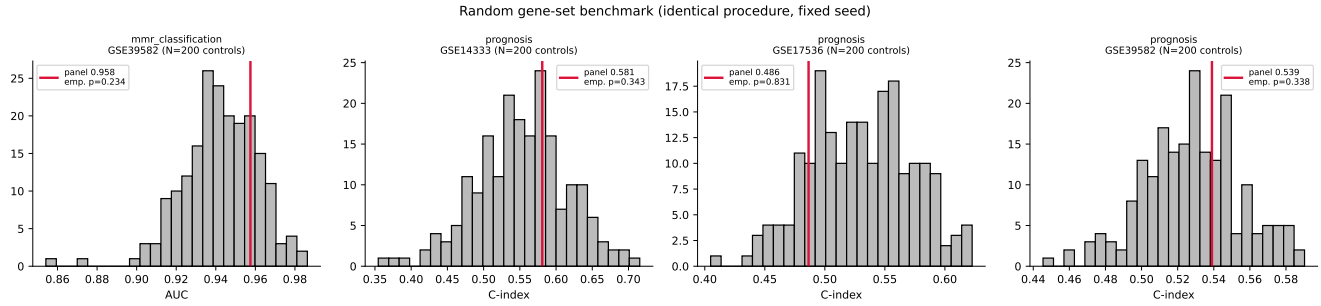


Figure 6: Panel performance (red line) versus the distribution of 200 random gene sets processed by the identical pipeline: dMMR classification (AUC) and frozen prognosis (C-index) in the three external cohorts.

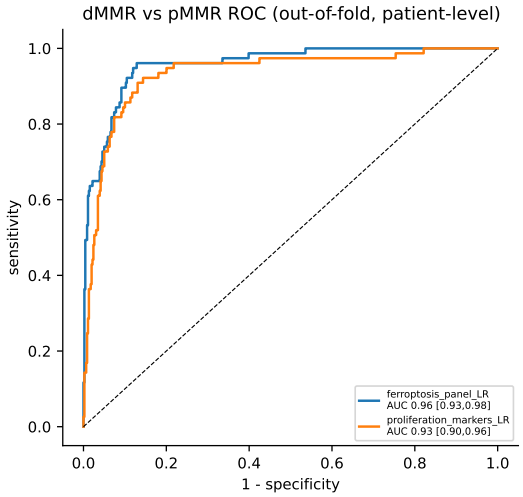


Figure 7: Out-of-fold ROC curves for dMMR vs. pMMR in GSE39582 (panel and proliferation markers).

Table 6: Positive controls for GPX4 dependence (59 lines, OOF AUC) against 200 random sets.

Signature	Genes	AUC	Random median	p_{emp}
Ferroptosis panel	36	0.760	0.776	0.637
EMT	24	0.590	0.755	0.980
Proliferation	12	0.671	0.722	0.771
Data-driven ceiling	36	0.818	–	–

Table 7: Ridge prediction of cell-line response from the panel (OOF Spearman ρ) versus random gene sets.

Response	n	ρ	Random med.	p_{emp}	q_{BH}
5-Fluorouracil	40	0.127	0.183	0.647	0.809
Irinotecan	40	−0.031	−0.108	0.388	0.809
Oxaliplatin	40	0.480	0.201	0.045	0.224
SN-38	39	−0.171	−0.146	0.557	0.809
GPX4 CRISPR effect	59	−0.054	0.334	0.985	0.985

log hazard ratio were 0.42, 0.64, 0.37 and 0.20. The high subtype contained fewer dMMR tumours in GSE39582 (3.1% vs. 15.9%, Fisher $p = 0.004$), but this difference was not larger than for random sets ($p_{\text{emp}} = 0.259$). No association with stage or age was found.

4.8 Cell-Line Sensitivity and Drug-Response Analysis

The panel predicted GPX4 dependence in 59 colorectal lines with OOF AUC 0.760 (95% CI 0.645–0.887; permutation $p = 0.010$), so the panel does contain information about the label, but random sets achieved a median AUC of 0.776 ($p_{\text{emp}} = 0.637$). The positive controls clarify the limits (Table 6, Fig. 9b): the EMT signature (AUC 0.590) and the proliferation signature (0.671) did not recover the expected signal, whereas selecting 36 genes inside folds gave a ceiling of 0.818. Hence, at $n = 59$, the benchmark has little power to separate a specific panel from random sets.

Projection of the frozen classifier onto tumours showed no association with survival in any cohort (HR per SD 0.99–1.24; all $p > 0.1$; p_{emp} 0.54–0.95), although the projected score correlated weakly with dMMR in GSE39582 (AUC 0.406, Mann–Whitney $p = 0.008$).

For chemotherapy response (Table 7), only oxaliplatin showed a correlation between predicted and measured ln IC_{50} exceeding random sets at the nominal level ($\rho = 0.480$, 95% CI 0.183–0.702; $p_{\text{emp}} = 0.045$), which is not significant after BH correction ($q = 0.224$). The other four models were not distinguishable from random sets. With five tests, and with the lower bound of p_{emp} limited by 200 sets, this should be regarded as a hypothesis for replication, not a finding.

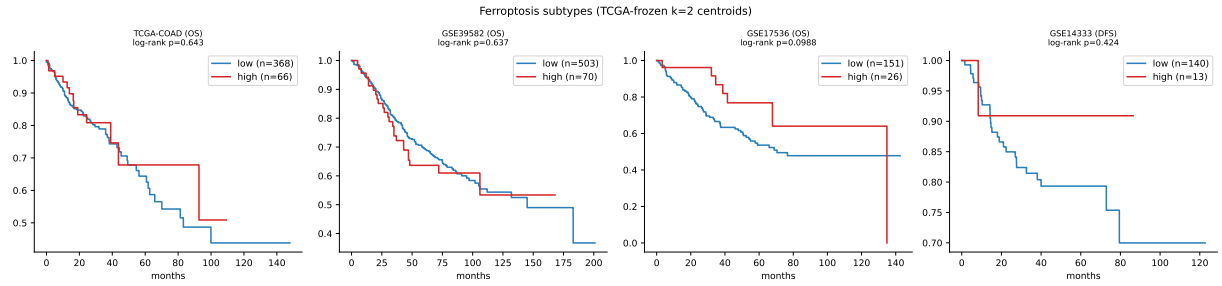


Figure 8: Kaplan–Meier curves for the two ferroptosis subtypes defined by TCGA-frozen centroids.

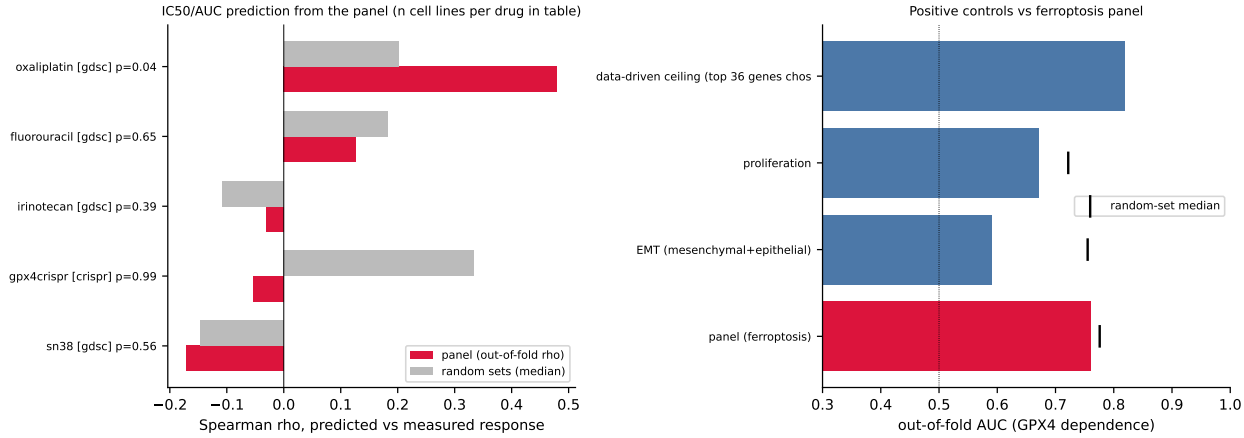


Figure 9: (a) Out-of-fold correlation between predicted and measured response for the panel (red) and median of random sets (grey). (b) Positive controls and ferroptosis panel for prediction of GPX4 dependence; black ticks mark random-set medians.

4.9 Sensitivity Analyses

Clinical stage was strongly prognostic in all cohorts (HR per SD 1.72–2.77), confirming that the survival machinery detects real signal. In contrast, proliferation (HR per SD 0.81–1.05) and EMT scores (1.10–1.22) had weak or inconsistent effects; EMT was nominally associated with OS in GSE39582 (HR 1.22, $p = 0.004$), but not larger than random sets ($p_{\text{emp}} = 0.075$).

4.10 Comparison with Published Baselines (2023 Onwards)

Table 8 sets the design of eleven studies published from 2023 onwards beside this study, and Table 9 compares the metrics that are of a similar type. Table 10 pairs each result of this study with its internal baseline.

Design-level patterns. Three patterns emerge from Table 8. First, GSE39582 is the external cohort in at least four of the eleven studies [13, 19, 20, 22], and the remaining studies mostly rely on TCGA for discovery. Second, colorectal models contain between 3 and 39 genes, and no gene set is shared among more than a few studies. Third, the comparators are clinical stage [13], alternative algorithms [12, 18] or nothing; only the pan-cancer benchmark [23] tests a random-gene null, in line with the concern raised earlier for expression signatures [16]. None of the colorectal studies describes the removal of patients duplicated between GEO series.

Metric-level reading. In GSE39582, the cohort shared with Wang [13], the published 3-year and 5-year AUCs of 0.62 and 0.63 exceed the frozen model by 0.06 to 0.08, and the bootstrap interval of the present estimate (0.499 to 0.614) lies at or below the published values. The strongest published value (random survival forest, 5-year AUC 0.965) was obtained in the discovery cohort and fell by about 0.32 in the single external dataset shown (Fig. 10), a loss expected when flexible learners are fitted and evaluated in the same data. The in-sample TCGA AUC of this study (0.616 and 0.634) is lower than the training AUCs of the 2026 model, which are optimistic in the same way. Published external AUCs (0.59 to 0.67 across horizons) cannot be judged against a null, because no colorectal study reports one. For orientation, the 95th percentile of random-set C-index in this study was 0.575 to 0.646 across the three external cohorts, although a different metric. The closest conceptual match is the pan-cancer benchmark [23], whose pan-cancer C-index of 0.573 lies inside the range of external C-indices found here (0.486 to 0.581), and whose integrated signature also failed to beat a simpler ferroptosis-only model. Wang additionally reported that the risk score did not separate predicted IC_{50} for 5-fluorouracil, irinotecan or oxaliplatin, in line with the absence of a robust drug-response signal here [13]. Accordingly, no claim is made that the present model outperforms or underperforms any published signature. The supported statement is narrower: external discrimination is modest and of the same order as published external values, and, unlike the colorectal studies, the share attributable to the specific genes is tested.

Table 8: Ferroptosis-related prognostic studies in colorectal cancer published from 2023 onwards and a pan-cancer benchmark, compared with this study. AUC is time-dependent ROC AUC. n.r.: not reported.

Study	Year	Signature	Discovery to validation	Reported performance	Baseline or null reported[†]
Guo [11]	2023	39 genes	TCGA and GEO, split not stated[a]	AUC 0.81, 0.81 and 0.78 at 2, 5 and 8 years	None described
Li [14]	2023	3 genes (SLC2A3, CDKN2A, FABP4)	TCGA to GEO	AUC 0.568 as re-evaluated in [13][b]	None described
Xu [17]	2023	5 genes from a ferroptosis-dependence score	Single-cell GSE161277 screening, then bulk cohorts	High predictive ability reported; qRT-PCR of signature genes	None described
Wang [18]	2023	23 genes, Lasso and plsRcox, recurrence in stage II and III	4 pooled datasets to 5 other datasets (1,397 samples)	Best of 83 algorithm combinations	83 model combinations compared
Zeng [19]	2024	4 genes (CYBB, PRNP, ACSL4, ACSL6), LASSO Cox	TCGA to GSE39582	AUC 0.897 (TCGA) and 0.858 (GSE39582) for the score plus 3 clinical factors	None described
Gan [12]	2024	6 intersection genes, random survival forest and nine other models	TCGA-COAD cross-validation to GSE29621 and other GEO sets	TCGA AUC 0.978, 0.985 and 0.965 at 1, 3 and 5 years; GSE29621 AUC 0.671, 0.589 and 0.649	Alternative survival models
Guan [20]	2024	10 genes (metabolic plasticity and ferroptosis), LASSO Cox	TCGA to GSE17536 and GSE39582	Nomogram combining signature and clinical factors	None described
Cuproptosis and ferroptosis [21]	2025	8 genes	TCGA and GEO pooled (n = 1,034), random 1:1 split	Time-dependent C-index	None described
Wang, mitochondrial [13]	2025	4 genes (PDSS2, GRSF1, SLC39A8, P4HA1), LASSO Cox	TCGA to GSE39582 (585 samples)	TCGA AUC 0.71, 0.75 and 0.74 at 3, 4 and 5 years; GSE39582 AUC 0.62, 0.64 and 0.63; nomogram C = 0.78	TNM stage, age and sex compared with the nomogram
Gao [22]	2026	8 genes (ALG3, CST1, RPS10, TRPV4, SPRR1A, FGFR4, TPM1, KDM1A)	TCGA (n = 630) to GSE72970 and GSE39582	Training AUC 0.857, 0.728 and 0.808 at 1, 3 and 5 years	Nomogram calibration only
Demir and Yasar [23]	2026	26 genes, three cell-death programmes	TCGA pan-cancer to METABRIC and CGGA	Pan-cancer C = 0.573 (0.552 to 0.594); adjusted HR 1.12 and 1.19	1,000 size-matched random gene sets; single-pathway models; proliferation adjustment
This study	2026	3 genes (ALOX12, AKR1C1, HSPB1), ridge Cox; 36-gene candidate panel	TCGA-COAD to GSE39582, GSE17536 and GSE14333 (de-duplicated)	External C 0.539, 0.486 and 0.581; 5-year AUC 0.553 in GSE39582	Age and stage model; 200 random 36-gene sets; positive controls

[†]Based on the abstracts, methods text and result summaries reviewed, not on re-analysis. "None described" means that no baseline or null model was found in the text reviewed. [a]Abstract only. [b]Second-hand value, not verified in the primary article. Reported metrics differ and are not directly comparable.

Table 9: Metric-level comparison of this study with published baselines that report a quantity of the same or a similar type. AUC and C-index are different quantities and are listed side by side but not ranked.

Baseline	Reported by the baseline	Closest metric of this study	Comparability
Wang 2025 [13]	GSE39582 AUC 0.62 (3 years) and 0.63 (5 years)	GSE39582 AUC 0.557 (3 years) and 0.553 (5 years), 95% interval 0.499 to 0.614	High: same metric type and same external cohort; estimators may differ
Gan 2024 [12]	Training AUC 0.965 and external AUC 0.649 at 5 years (GSE29621)	TCGA in-sample AUC 0.634 and GSE39582 AUC 0.553 at 5 years	Moderate: same metric type, different external cohort
Gao 2026 [22]	Training AUC 0.728 (3 years) and 0.808 (5 years)	TCGA in-sample AUC 0.616 (3 years) and 0.634 (5 years)	Moderate: both values are in-sample and optimistic
Demir and Yasar 2026 [23]	Pan-cancer C 0.573; integrated signature beat size-matched random sets (empirical $p < 0.001$)	External C 0.486 to 0.581; empirical p 0.338, 0.831 and 0.343 against 200 random sets	Moderate: same null concept and metric type, but a different cancer scope
Zeng 2024 [19]	AUC 0.858 in GSE39582 for the score plus 3 clinical factors	C 0.683 for age, stage and risk score in GSE39582; score alone 0.539	Low: AUC versus C-index, combined model, no clinical-only baseline reported
Other studies in Table 8	No externally validated metric of a comparable type in the text reviewed	Not applicable	Not compared quantitatively

Genes shared with an earlier signature. The three genes selected by TCGA screening include AKR1C1 and ALOX12, which also appear in a five-gene colon signature published in 2021 [15]. That model was published before the 2023 window and is therefore not part of Table 8. It was reported to reach AUCs of 0.546 to 0.578 at 1 to 3 years in its GEO test cohort, close to the external discrimination observed here, which suggests that these genes carry only weak prognostic information outside TCGA.

Caveats. Cohorts, endpoints, gene-set definitions and AUC estimators differ between studies, so the comparison is indicative and not a formal meta-analysis. Published models were not refitted. Several entries were taken from abstracts, and

Table 10: Summary of each result of this study against its internal baseline.

Question	Score or model	Baseline	Outcome
Prognosis beyond clinical data	Frozen molecular model, C 0.539, 0.486 and 0.581	Age and stage model, C 0.675, 0.747 and 0.624	Molecular model below the clinical model in all three cohorts
Specificity of prognosis	Empirical p 0.338, 0.831 and 0.343	200 random 36-gene sets, median C 0.528, 0.532 and 0.555	Not distinguishable from random sets
Survival of subtypes	High versus low subtype, log-rank p 0.643, 0.637, 0.099 and 0.424	200 random sets, empirical p 0.42, 0.64, 0.37 and 0.20	Not distinguishable from random sets
dMMR classification	Panel AUC 0.958	200 random sets (empirical p 0.234); proliferation markers AUC 0.934	Not distinguishable from random sets
GPX4 dependence in cell lines	Panel AUC 0.760, permutation p 0.010	Random-set median AUC 0.776 (empirical p 0.637); EMT 0.590; proliferation 0.671; ceiling 0.818	Signal present against label permutation, not specific to the panel
Oxaliplatin IC_{50}	Out-of-fold ρ 0.480	Random-set median ρ 0.201 (empirical p 0.045, BH q 0.224)	Nominal only; not significant after correction

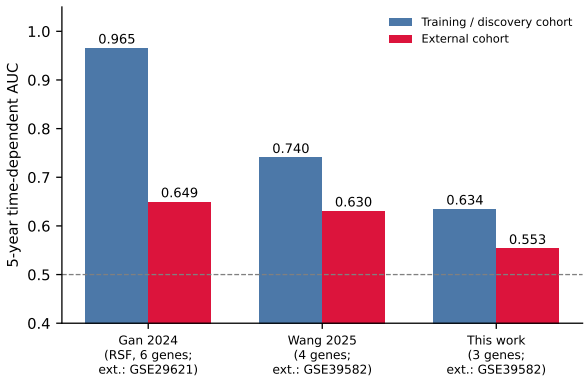


Figure 10: Five-year time-dependent AUC in discovery and external cohorts for two published ferroptosis models and this study. The dashed line marks chance. For this study the discovery value is the TCGA in-sample AUC (optimistic) and the external value is GSE39582.

the entries for Xu, Wang 2023, Guan, the cuproptosis and ferroptosis model, and Gao were not re-verified against the full texts.

5 Discussion

A weak and non-replicated prognostic signal. The frozen three-gene score was not significantly associated with outcome in any single external cohort, and the pooled OS hazard ratio per SD of 1.09 (0.97 to 1.22) included 1. The adjusted estimates disagreed between the two OS cohorts, and the increment in C-index over age and stage ranged from −0.022 to +0.018. Discrimination of the score alone (0.49 to 0.58) was close to chance, whereas age and stage reached 0.62 to 0.75. The in-sample TCGA value (0.617) and the cross-validated value (0.548) differ by almost as much as the external values differ from chance, which is the usual signature of optimism in a discovery cohort. Only ALOX12 survived multiplicity correction in the TCGA screening, so the signal available for model building was thin from the start.

No gene-specific information. In none of the three external cohorts did the panel outperform random gene sets of the same size (empirical p 0.338, 0.831 and 0.343), and the a priori FerroScore, which needs no fitting, had a pooled OS hazard ratio of 0.95 (0.86 to 1.05). The one nominal component result (driver mean in GSE39582) was not replicated and arises among twelve tests. These data therefore do not support the idea that the annotated ferroptosis role of the genes carries prognostic information in colon cancer. The proliferation markers were prognostic in one cohort only (C = 0.570 in GSE39582, 0.498 and 0.509 elsewhere), so no gene set was consistently informative.

Mismatch repair status. An AUC of 0.958 for dMMR would conventionally be read as strong evidence. Random gene sets of equal size reached similar values (p_{emp} = 0.234), and the unsupervised driver-minus-suppressor score had AUC 0.453. The result reflects the broad transcriptomic footprint of mismatch-repair deficiency and not ferroptosis biology. The same applies to the ferroptosis subtypes, whose lower dMMR fraction in GSE39582 (3.1% against 15.9%) was not larger than for random sets.

Cell-line sensitivity and drug response. The panel predicted GPX4 dependence with an AUC of 0.760 that was significant against label permutation (p = 0.010) but unremarkable against random sets (median 0.776). A permutation test answers whether any signal exists, whereas the random-set test answers whether the signal belongs to the gene list, and only the second matters for a signature. The positive controls limit what can be concluded: the EMT signature (0.590) and the proliferation signature (0.671) did not recover the expected signal, while the in-fold ceiling reached 0.818, so the benchmark has limited power with 59 lines. Oxaliplatin was the only drug model that exceeded random sets nominally (ρ = 0.480, p_{emp} = 0.045), and it did not survive correction (q = 0.224). It is a hypothesis for replication and not a finding. Cell-line response is not a clinical dose.

Results against baselines. Table 10 collects every result beside its baseline. The pattern is consistent: whenever a baseline is available, the ferroptosis panel is either below it (clinical model), indistinguishable from it (random gene sets), or supported only nominally (oxaliplatin). Two checks show that the pipeline is able to detect signal when it exists. Clinical stage was strongly prognostic in every cohort (hazard ratio per SD 1.72 to 2.77), and the data-driven ceiling in cell lines was clearly above the panel.

Comparison with recent studies. Table 8 and Table 9 place the results beside ferroptosis models published from 2023 onwards. The published external AUCs in GSE39582 (0.62 to 0.64) are somewhat higher than the 0.553 to 0.557 of the frozen model, but the published values come from models without a random-gene null, and the largest published performances (AUC up to 0.985) were obtained in discovery cohorts. The only study that applied a random-gene null reached the same qualitative conclusion for an integrated signature. The supported statement is narrower than any claim of superiority or inferiority: external discrimination is modest and of the same order as published external values, and the present design tests how much of it belongs to the specific genes.

Practical implications. The scores in this study do not support clinical use for prognosis beyond stage or for treatment selection. The results support random-signature benchmarking, frozen external validation, de-duplication of overlapping GEO cohorts and positive controls as routine practice, because two GEO series in this work shared 95 samples that would otherwise have inflated the evidence.

6 Limitations

The gene panel is a literature-derived candidate list that has not been curated against a ferroptosis database, and findings may differ for another panel. GSE14333 provides DFS only and has 29 events after de-duplication, so the third validation is exploratory. The adjusted analysis in GSE17536 rests on 177 patients and produced an estimate in the opposite direction, which shows the instability of small-cohort adjustment. The component analysis and the tertile analysis were added after the main results and are exploratory, and many secondary tests report nominal p values. Cell-line analyses use 39 to 59 lines, the retrieved GDSC2 file contained no GPX4 inhibitors so that GPX4 CRISPR dependence served as a surrogate, CTRP could not be retrieved, and the power to separate the panel from random sets is correspondingly limited, as the positive controls show. TCGA microsatellite status and DFS were not retrieved, so dMMR classification was run only in GSE39582, with labels parsed from GEO annotations. RNA-seq and microarray data are not pooled, and cohort-wise standardisation assumes comparable case mix. Empirical p values cannot fall below one in 201. Published signatures were not refitted, so the comparison with them concerns design and reported metrics, and several entries were taken from abstracts. No adjustment for proliferation or stroma was made for the frozen score. Retrospective data carry selection bias and residual confounding, and no clinical application is supported without experimental and prospective validation.

7 Conclusion

This study presents a reproducible pipeline that pairs every ferroptosis-signature result with a baseline or null model. In colon cancer, a 36-gene ferroptosis panel did not outperform random gene sets for external survival prediction, dMMR classification, subtype survival association or GPX4-dependence prediction. The frozen three-gene score was not significantly associated with outcome in any single external cohort, added little or negative discrimination to age and stage, and the a priori FerroScore was null. A possible oxaliplatin signal merits replication. Compared with ferroptosis models published from 2023 onwards, external discrimination was of a similar order, whereas the colorectal studies did not report a random-gene null. Future work should use a curated panel, GPX4-inhibitor response data and larger cell-line panels, and should adjust for proliferation and stroma.

Declarations

Ethics approval: Not applicable; only public, de-identified data were analysed.

Data availability: TCGA-COAD (GDC), GEO accessions GSE39582, GSE17536 and GSE14333, GDSC2 and DepMap are public.

Code availability: `ferro_pipeline.py` v1.5.0; add repository URL before submission.

Funding: [Declare funding.]

Conflicts of interest: [Declare any competing interests.]

Author contributions: [Specify contributions.]

A Gene Panel

Table 11: The 36-gene candidate ferroptosis panel.

Role	Genes
Driver (16)	ACSL4, LPCAT3, ALOX15, ALOX12, ALOX5, POR, PEBP1, NCOA4, TFRC, HMOX1, SAT1, NOX1, TP53, GLS2, DPP4, BECN1
Suppressor (20)	GPX4, SLC7A11, AIFM2, GCH1, NFE2L2, FTH1, FTL, SLC3A2, HSPB1, ACSL3, SCD, DHODH, NQO1, GCLC, GCLM, TXNRD1, SLC40A1, CISD1, FANCD2, AKR1C1

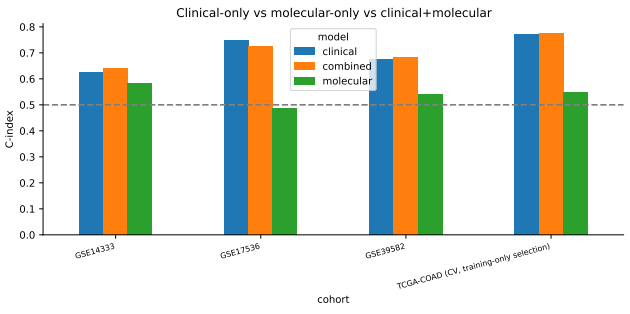


Figure 11: Clinical-only, molecular-only and combined model discrimination by cohort.

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